

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

```
df = pd.read_csv("PlayGolf-MissingData.csv")
```

```
df.head(20)
```

	Unnamed: 0	Date	Weekday	Holiday	Temp	Humidity	Wind	Outlook	Crowdedness
0	0	08-01	0	0.0	25.1	99.0	0.0	rainy	0.14
1	1	08-02	1	0.0	26.4	NaN	0.0	sunny	NaN
2	2	08-03	2	0.0	NaN	96.0	0.0	rainy	0.21
3	3	08-04	3	0.0	24.1	68.0	0.0	overcast	0.68
4	4	08-05	4	NaN	24.7	98.0	0.0	rainy	0.20
5	5	08-06	5	0.0	26.5	98.0	0.0	NaN	0.32
6	6	08-07	6	0.0	27.6	78.0	0.0	rainy	0.72
7	7	08-08	0	0.0	28.2	NaN	0.0	rainy	0.61
8	8	08-09	1	0.0	27.1	70.0	0.0	overcast	NaN
9	9	08-10	2	1.0	26.7	75.0	NaN	sunny	0.54
10	10	08-11	3	0.0	NaN	NaN	0.0	NaN	NaN
11	11	08-12	4	NaN	24.3	77.0	0.0	overcast	0.67
12	12	08-13	5	0.0	23.1	77.0	1.0	sunny	0.66
13	13	08-14	6	0.0	22.4	89.0	1.0	rainy	0.38
14	14	08-15	0	0.0	NaN	80.0	1.0	sunny	0.46
15	15	08-16	1	0.0	26.5	88.0	1.0	rainy	NaN
16	16	08-17	2	0.0	28.6	76.0	0.0	NaN	0.52
17	17	08-18	3	0.0	NaN	NaN	NaN	rainy	NaN
18	18	08-19	4	0.0	27.0	73.0	1.0	overcast	0.62
19	19	08-20	5	0.0	26.9	73.0	0.0	sunny	0.81

```
df.shape
```

```
(20, 9)
```

```
df.columns
```

```
Index(['Unnamed: 0', 'Date', 'Weekday', 'Holiday', 'Temp', 'Humidity', 'Wind',
      'Outlook', 'Crowdedness'],
      dtype='object')
```

```
df.describe()
```

	Unnamed: 0	Weekday	Holiday	Temp	Humidity	Wind	Crowdedness
count	20.00000	20.000000	18.000000	16.000000	16.000000	18.000000	15.000000
mean	9.50000	2.850000	0.055556	25.950000	82.187500	0.277778	0.502667
std	5.91608	1.980829	0.235702	1.801111	10.765493	0.460889	0.209061
min	0.00000	0.000000	0.000000	22.400000	68.000000	0.000000	0.140000
25%	4.75000	1.000000	0.000000	24.600000	74.500000	0.000000	0.350000
50%	9.50000	3.000000	0.000000	26.500000	77.500000	0.000000	0.540000
75%	14.25000	4.250000	0.000000	27.025000	90.750000	0.750000	0.665000
max	19.00000	6.000000	1.000000	28.600000	99.000000	1.000000	0.810000

```
df = df.drop(columns="Unnamed: 0")
```

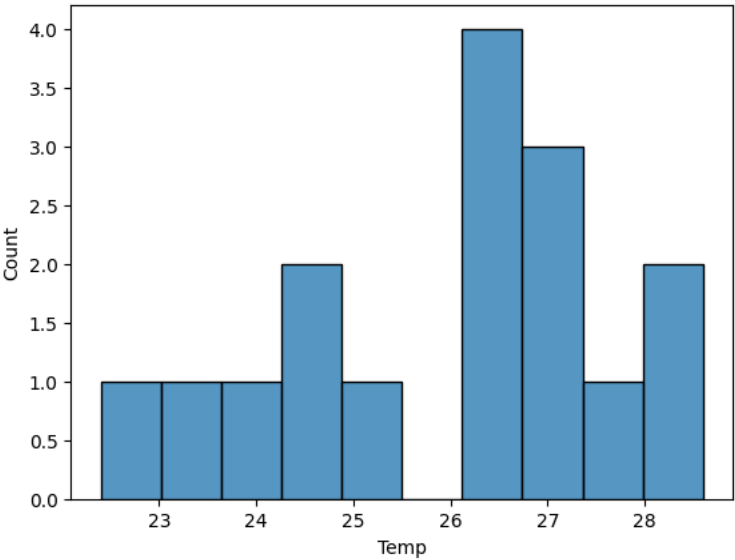
```
df.isnull().sum()
```

	0
Date	0
Weekday	0
Holiday	2
Temp	4
Humidity	4
Wind	2
Outlook	3
Crowdedness	5

```
dtype: int64
```

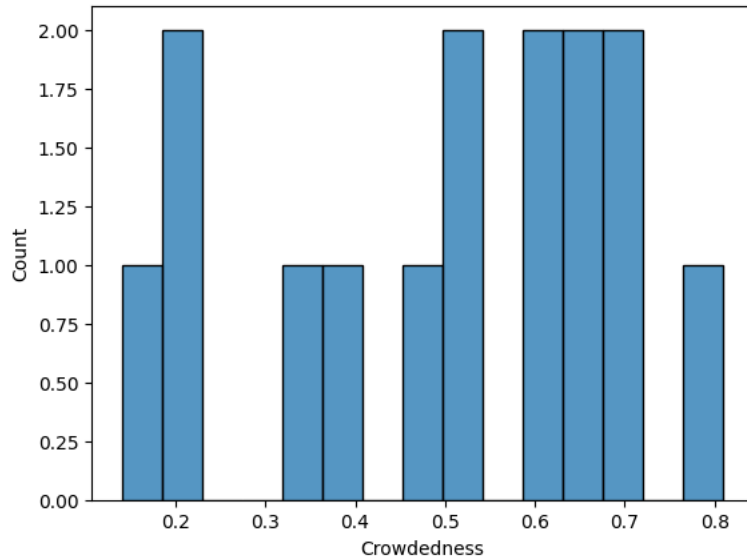
```
sns.histplot(df["Temp"], bins=10)
```

<Axes: xlabel='Temp', ylabel='Count'>



```
sns.histplot(df["Crowdedness"], bins=15)
```

<Axes: xlabel='Crowdedness', ylabel='Count'>



```
df_clean = df[df.isnull().sum(axis=1) < 4].reset_index(drop=True).copy()
```

```
df_clean.head()
```

	Date	Weekday	Holiday	Temp	Humidity	Wind	Outlook	Crowdedness
0	08-01	0	0.0	25.1	99.0	0.0	rainy	0.14
1	08-02	1	0.0	26.4	NaN	0.0	sunny	NaN
2	08-03	2	0.0	NaN	96.0	0.0	rainy	0.21
3	08-04	3	0.0	24.1	68.0	0.0	overcast	0.68
4	08-05	4	NaN	24.7	98.0	0.0	rainy	0.20

```
df_clean["Humidity"] = df_clean["Humidity"].fillna(df_clean["Humidity"].mean())
df_clean["Holiday"] = df_clean["Humidity"].fillna(df_clean["Holiday"].mode()[0])
```

```
df_clean.head()
```

	Date	Weekday	Holiday	Temp	Humidity	Wind	Outlook	Crowdedness
0	08-01	0	99.0000	25.10	99.0000	0.0	rainy	0.14
1	08-02	1	82.1875	26.40	82.1875	0.0	sunny	NaN
2	08-03	2	96.0000	25.25	96.0000	0.0	rainy	0.21
3	08-04	3	68.0000	24.10	68.0000	0.0	overcast	0.68
4	08-05	4	98.0000	24.70	98.0000	0.0	rainy	0.20

```
df_clean["Temp"] = df_clean["Temp"].interpolate(method="linear")
df_clean.head()
```

	Date	Weekday	Holiday	Temp	Humidity	Wind	Outlook	Crowdedness
0	08-01	0	99.0000	25.10	99.0000	0.0	rainy	0.14
1	08-02	1	82.1875	26.40	82.1875	0.0	sunny	NaN
2	08-03	2	96.0000	25.25	96.0000	0.0	rainy	0.21
3	08-04	3	68.0000	24.10	68.0000	0.0	overcast	0.68
4	08-05	4	98.0000	24.70	98.0000	0.0	rainy	0.20

```
df_clean["Outlook"] = df_clean["Outlook"].ffill()
df_clean.head()
```

	Date	Weekday	Holiday	Temp	Humidity	Wind	Outlook	Crowdedness
0	08-01	0	99.0000	25.10	99.0000	0.0	rainy	0.14
1	08-02	1	82.1875	26.40	82.1875	0.0	sunny	NaN
2	08-03	2	96.0000	25.25	96.0000	0.0	rainy	0.21
3	08-04	3	68.0000	24.10	68.0000	0.0	overcast	0.68
4	08-05	4	98.0000	24.70	98.0000	0.0	rainy	0.20

df_clean.isnull().sum()

	0
Date	0
Weekday	0
Holiday	0
Temp	0
Humidity	0
Wind	1
Outlook	0
Crowdedness	3

dtype: int64